

4
(5)

Measures of Central Tendency

The Mode

Measures of Central Tendency (The Mode)

- The mode is the number that appears *most frequently* in a set. A set of numbers may have one mode, more than one mode, or no mode at all.

Definition 1:- If the data set has only one mode, then we call it a *uni-modal* data.

Measures of Central Tendency (The Mode)

Definition 2:- If the data set has two modes, then we call it a *bi-modal* data.

Definition 3:- If the data set has more than two modes, then we call it a *multi-modal* data.

Measures of Central Tendency (The Mode)

Example 1:- Find the mode for each of the following observations.

i) 3, **5**, 9, **5**, 6, 2, 1 – Uni-modal

ii) **8**, 2, **8**, 3, 5, 7, **8**, 7 – Uni-modal

iii) 7, 1, 8, 2, 19 – No mode

Measures of Central Tendency (The Mode)

iv) 6, 8, 0, 7, 0, 3, 2 – Uni-modal

v) 4, 8, 4, 6, 4, 1, 8, 8, 9 – Bi-modal

vi) 2, 6, 3, 6, 5, 1, 1, 3, 8, 7 – Multi-modal

Measures of Central Tendency (The Mode)

Since the definition of mode is the observation that occurs most often, in the case of the GFD, the mode is assumed to be found in that class with the *highest frequency which is called the modal class.*

To find the mode in the case of GFD, we shall use the following formula.

Measures of Central Tendency (The Mode)

$$\hat{x} = \text{LCB} + \left[\frac{f_1 - f_0}{(f_1 - f_0) + (f_1 - f_2)} \right] \times W$$

where f_1 – the frequency of the modal class.

f_0 – the frequency of the class which is just before the modal class.

f_2 – the frequency of the class which is just after the modal class.

w – the class width of the GFD

Measures of Central Tendency (The Mode)

$$\hat{x} = \text{LCB} + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \cdot W$$

where f_1 – the frequency of the modal class.

f_0 – the frequency of the class which is just before the modal class.

f_2 – the frequency of the class which is just after the modal class.

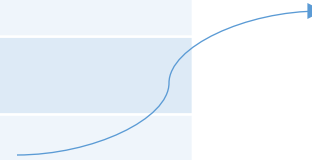
w – the class width of the GFD

Measures of Central Tendency (The Mode)

Example 2:- The following GFD refers to the weight (to the nearest Kg) of a sample of students at HiLCoE. Find the modal weight and interpret your result.

Weight (in Kg)	Number of students
35 - 43	8
44 - 52	11
53 - 61	16
62 - 70	19
71 - 79	14
80 - 88	9

Modal class



Measures of Central Tendency (The Mode)

As we can see from the table, the 4th class is the modal class. Thus

$$\begin{aligned}\hat{x} &= \text{LCB} + \left[\frac{f_1 - f_0}{(f_1 - f_0) + (f_1 - f_2)} \right] \times W \\ &= 61.5 + \left[\frac{19 - 16}{(19 - 16) + (19 - 14)} \right] \times 9 = 61.5 + \left[\frac{3}{3 + 5} \right] \times 9 \\ &= 61.5 + \left[\frac{27}{8} \right] = 61.5 + 3.375 \approx 64.9\end{aligned}$$

Measures of Central Tendency (The Mode)

Interpretation:- *Of all the students in the sample, those whose weight is 64.9 were large in number.*

Measures of Central Tendency (The Mode)

Advantages and Disadvantages of the Mode:

Measures of Central Tendency (The Mode)

Advantages:

- The mode is easy to understand and calculate.
- The mode is not affected by extreme values.
- The mode is easy to identify in un-grouped data and discrete frequency distribution.

Measures of Central Tendency (The Mode)

- The mode is useful for qualitative data.
- The mode can be located graphically.

Disadvantages:

- The mode is not well defined.
- The mode is not based on all values.
- The mode is stable for large values and will not be well defined if the data consist of a small number of values.

Measures of Central Tendency (The Mode)

- The mode is not capable of further mathematical treatment.
- Sometimes data have one mode, more than one mode, or no mode at all.

Relationship between the Mean, the Median and the Mode

MCT (Symmetric and Skewed distributions)

In statistics distributions are usually divided in to two measure groups referred to their mean.

If one tail is longer than another, the distribution is *skewed*.

MCT (Symmetric and Skewed distributions)

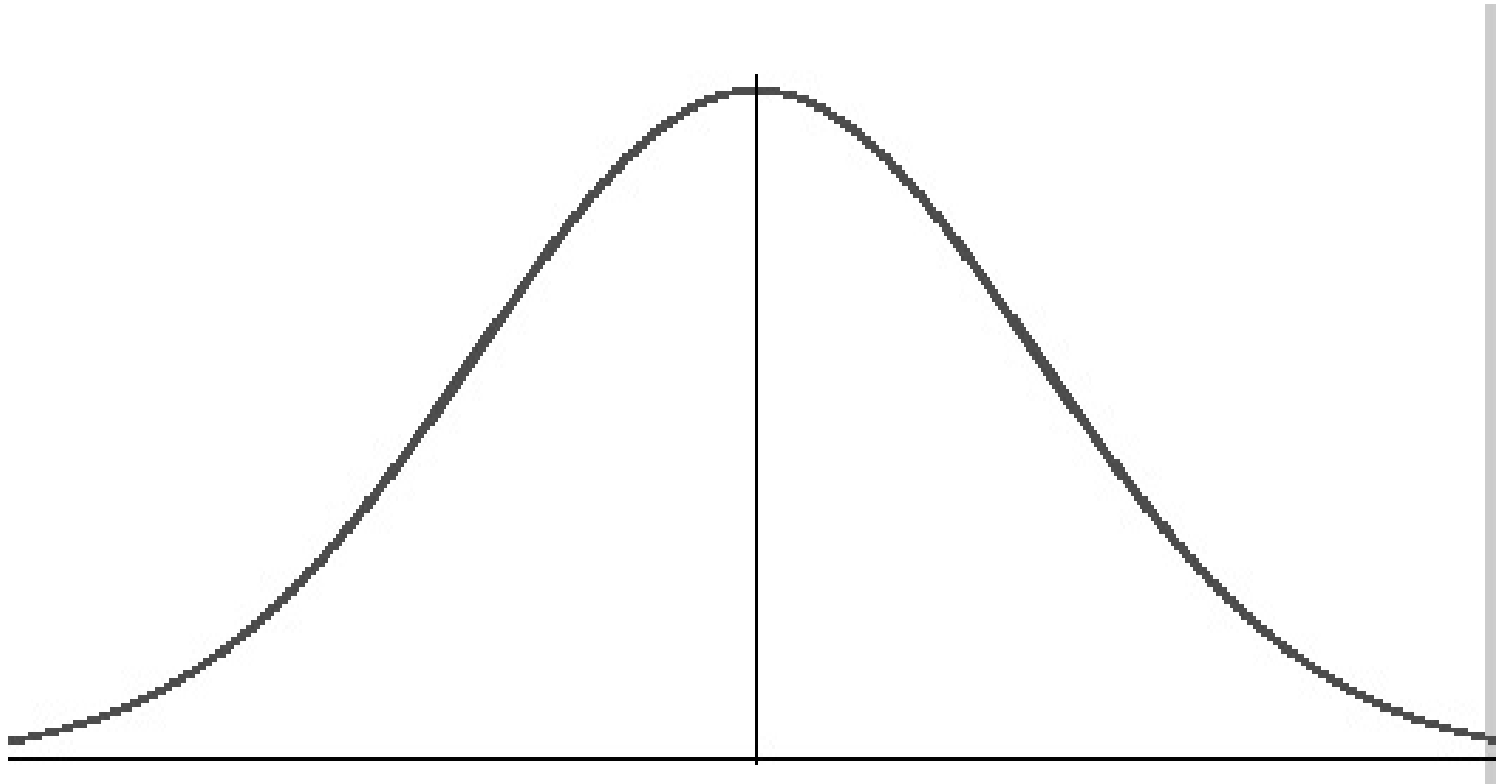
These distributions are sometimes called *asymmetric* or *asymmetrical distributions* as they don't show any kind of symmetry.

Symmetry means that one half of the distribution is a mirror image of the other half.

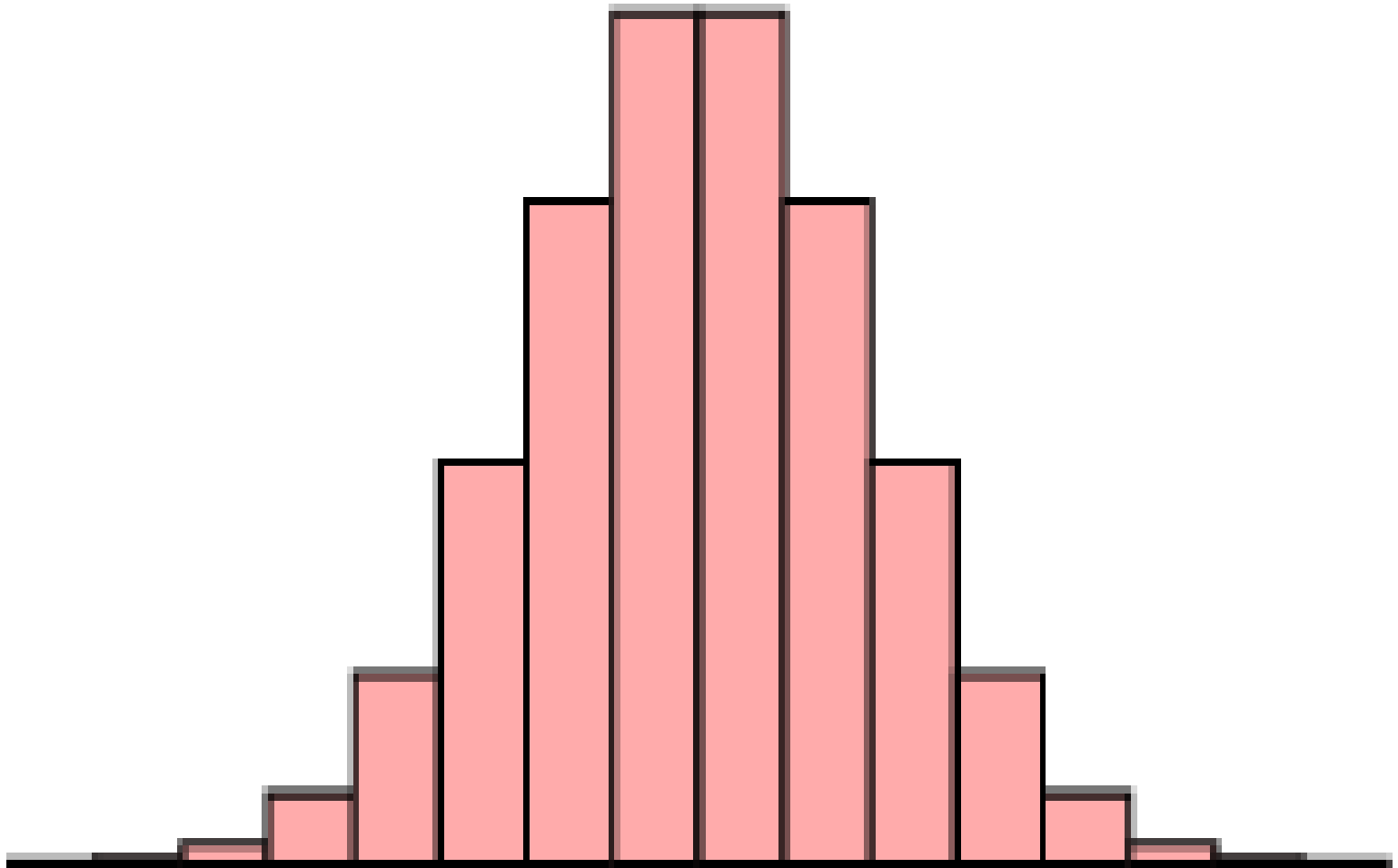
MCT (Symmetric and Skewed distributions)

For example, the *normal distribution* is a *symmetric distribution* with no skew.

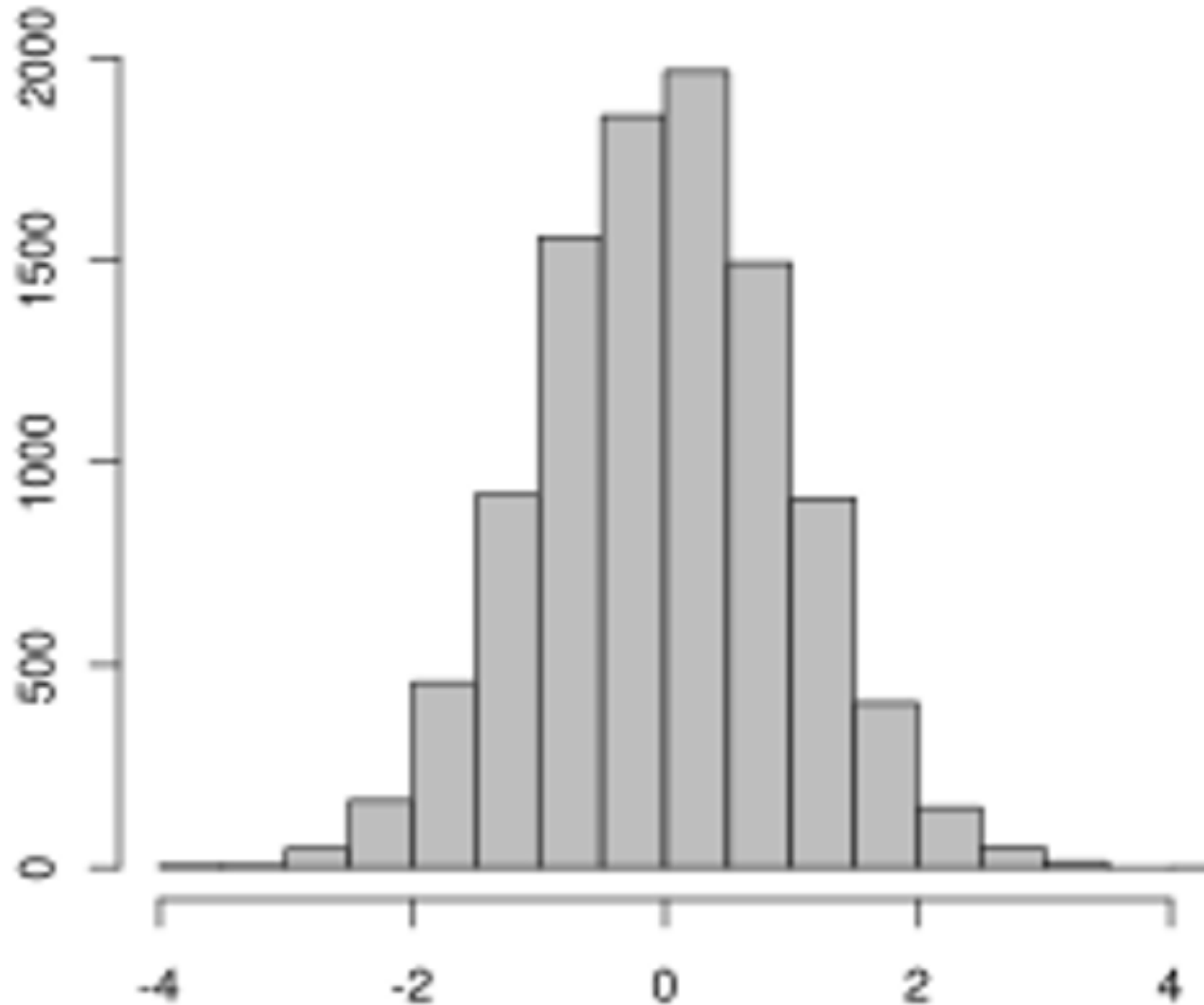
The tails are exactly the same.



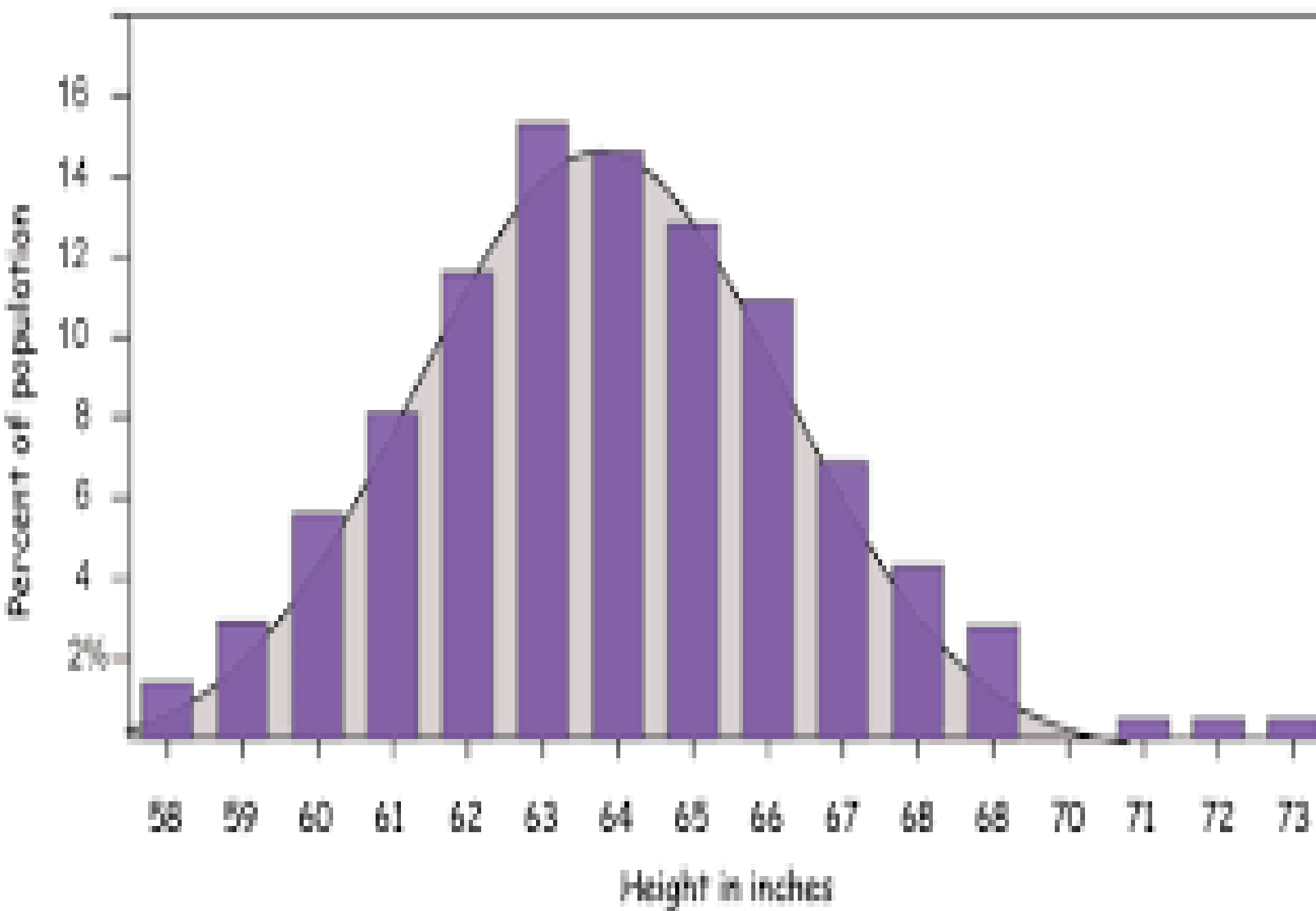
MCT (Symmetric and Skewed distributions)



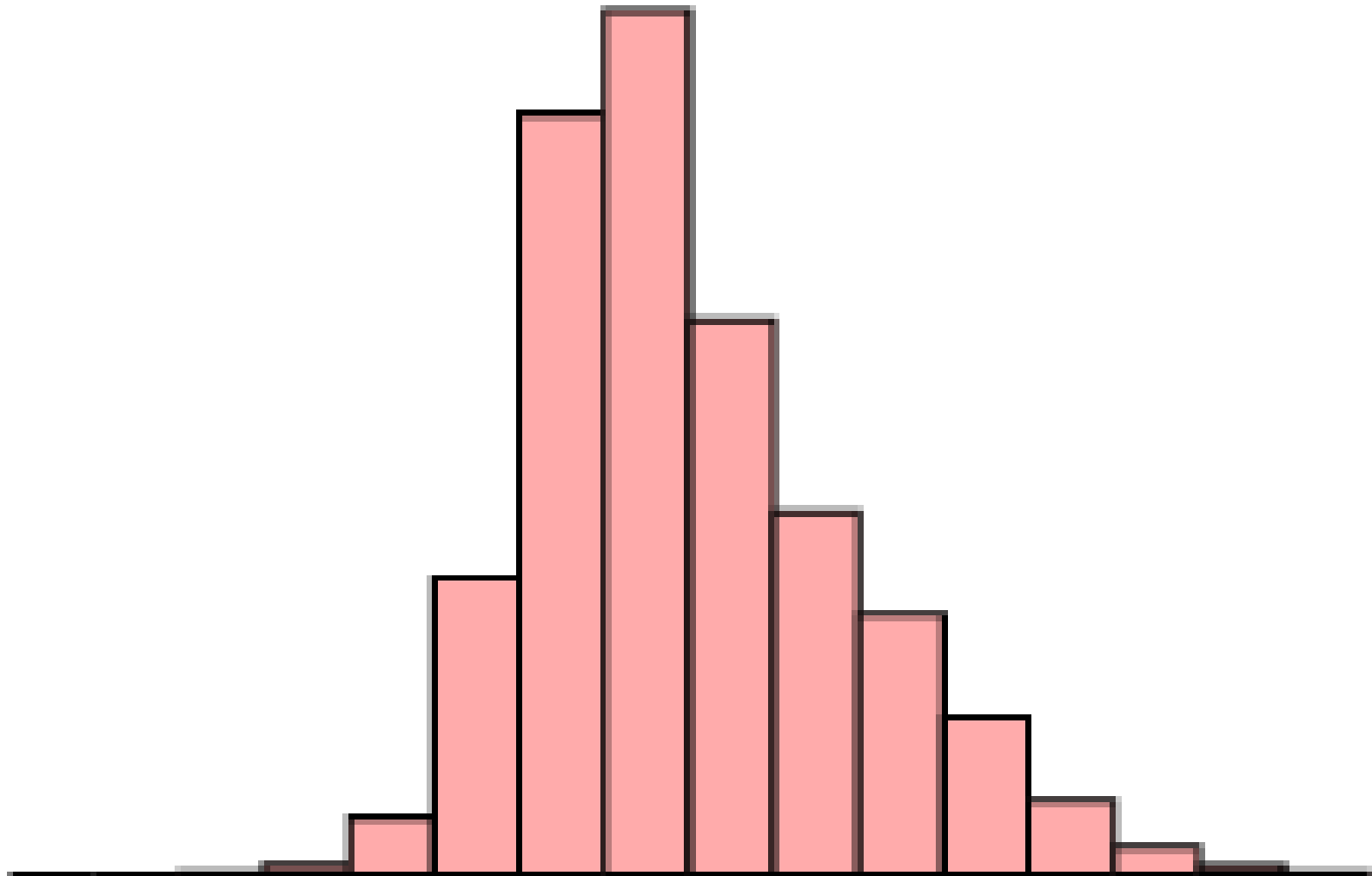
MCT (Symmetric and Skewed distributions)



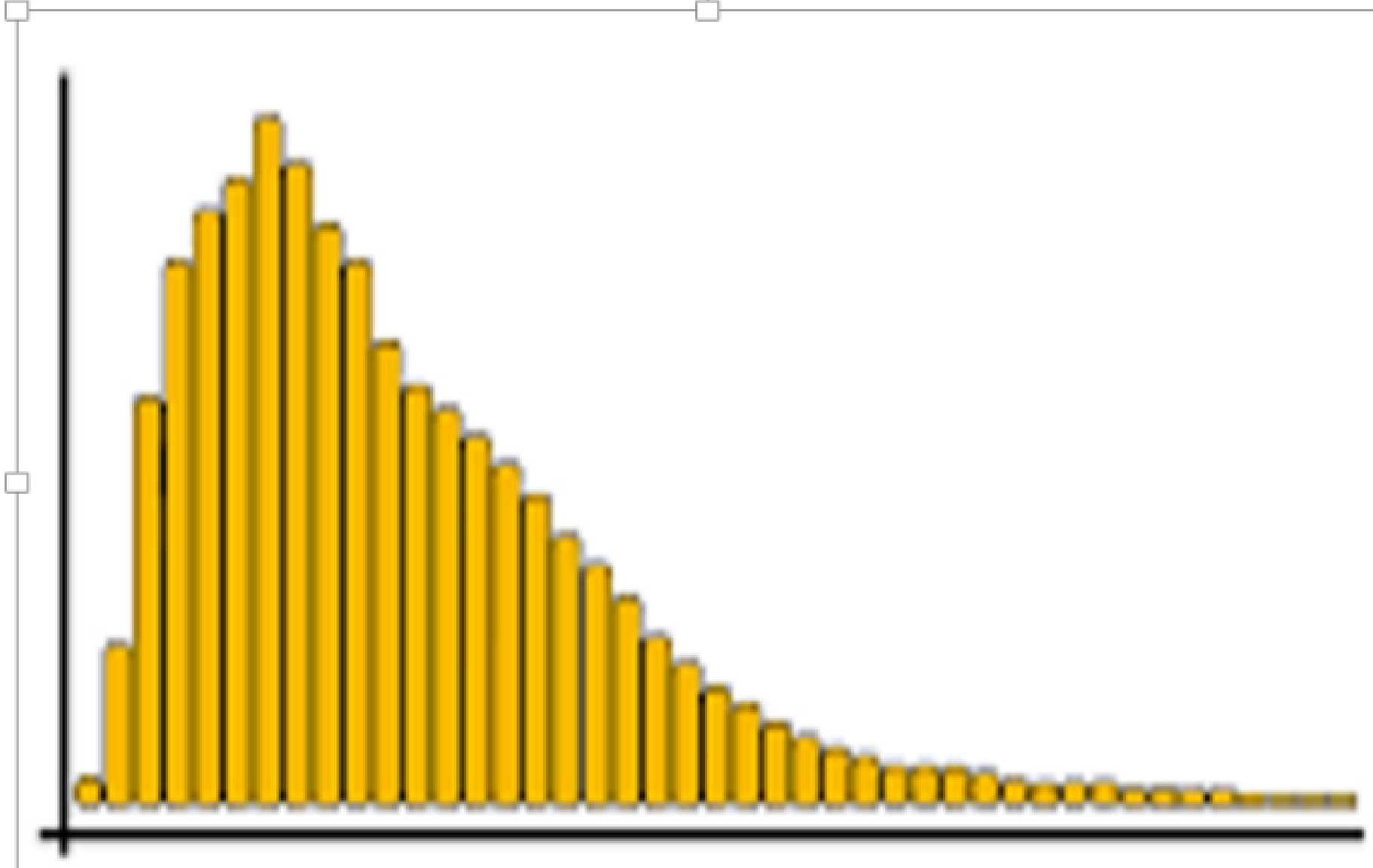
Heights (in.) of American women (ages 30-39)



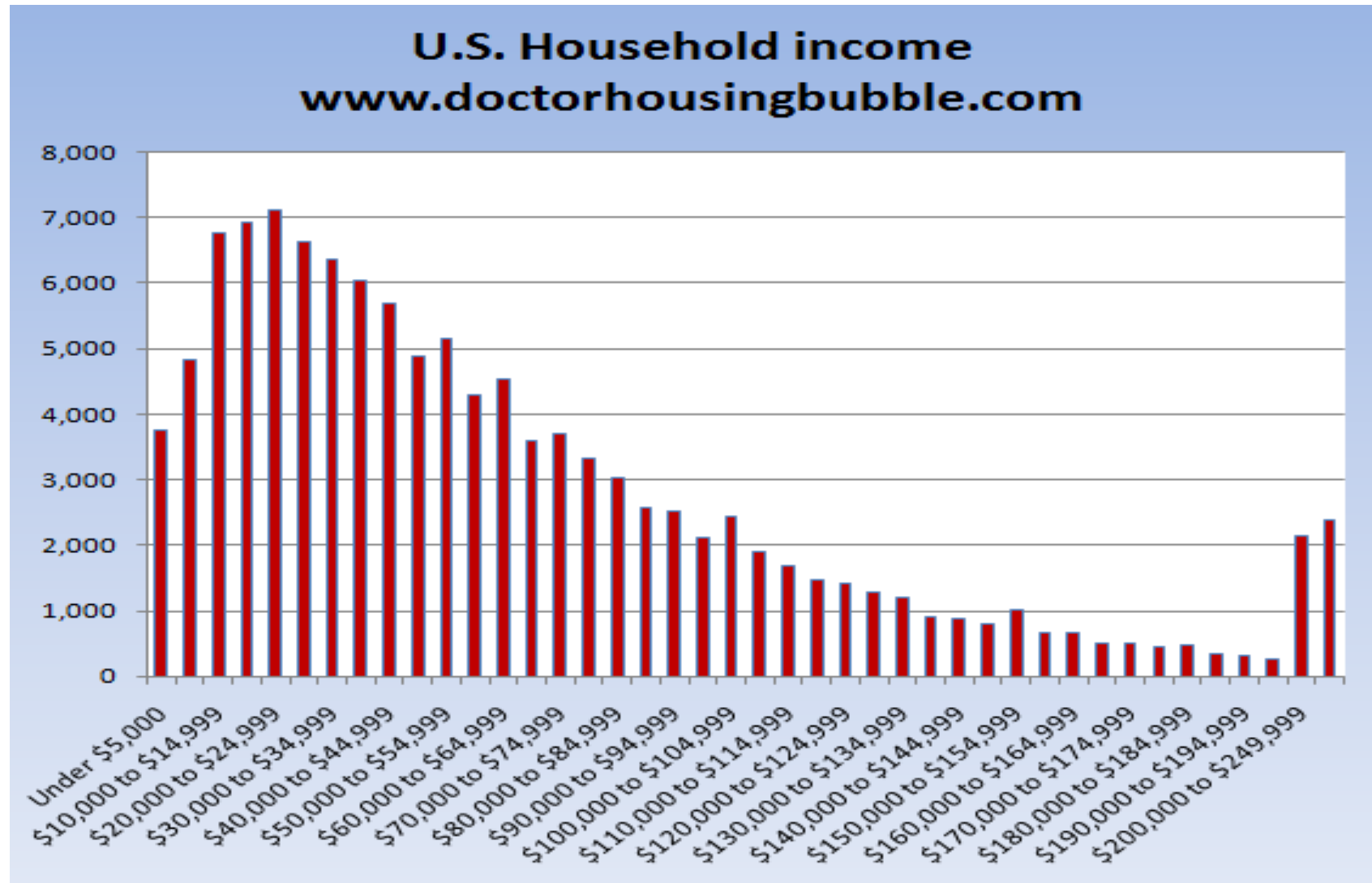
MCT (Symmetric and Skewed distributions)



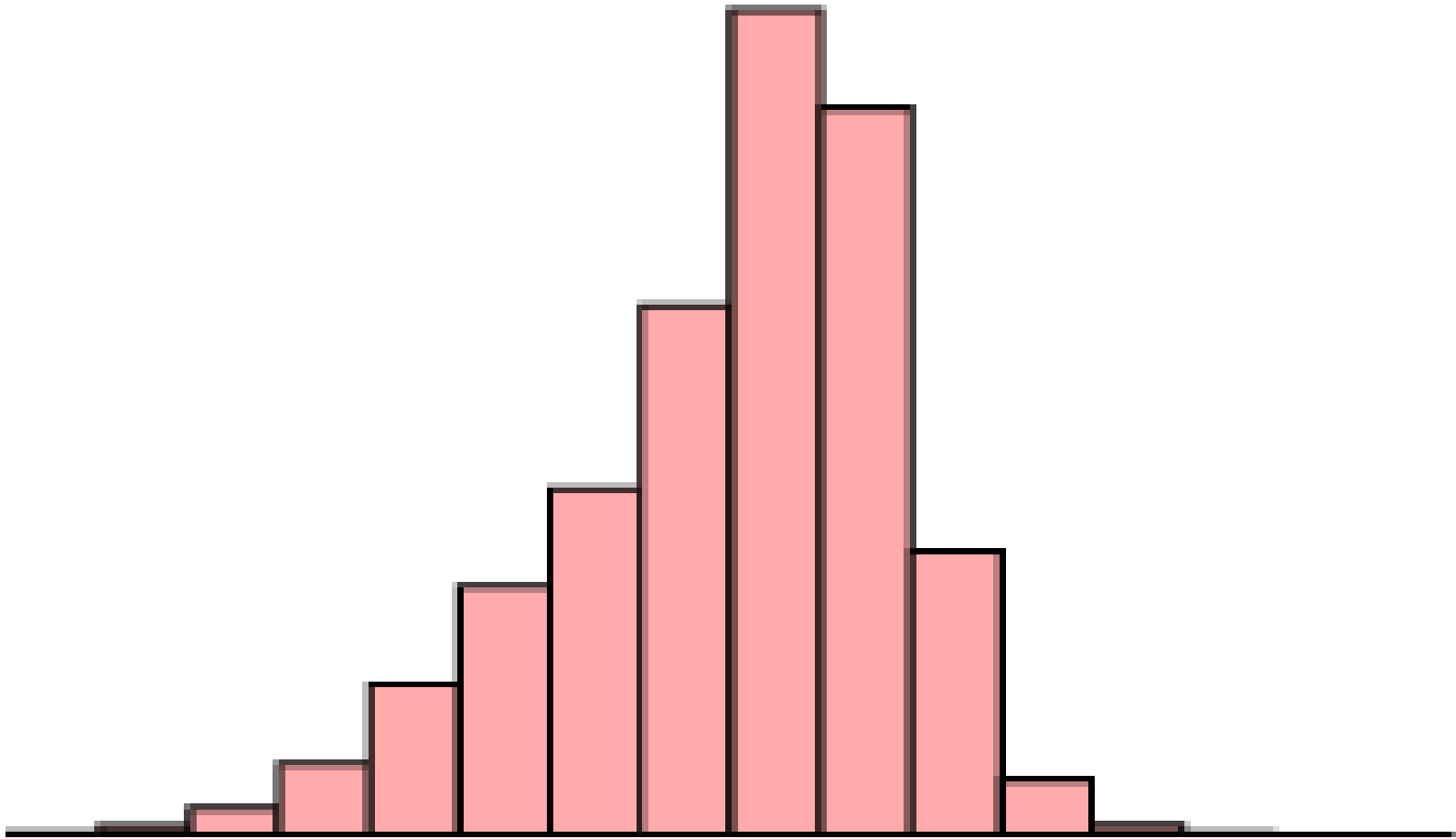
MCT (Symmetric and Skewed distributions)



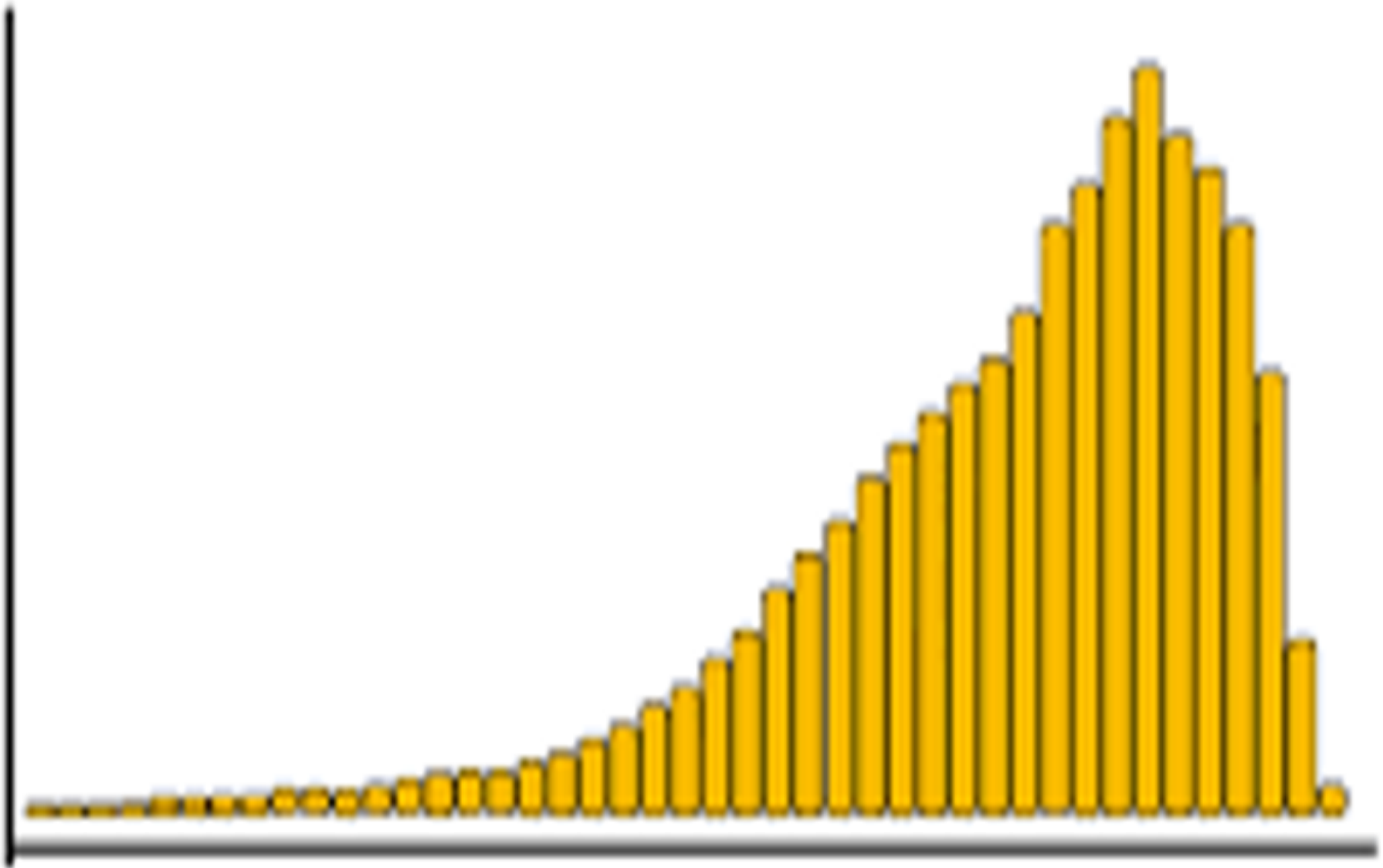
MCT (Symmetric and Skewed distributions)



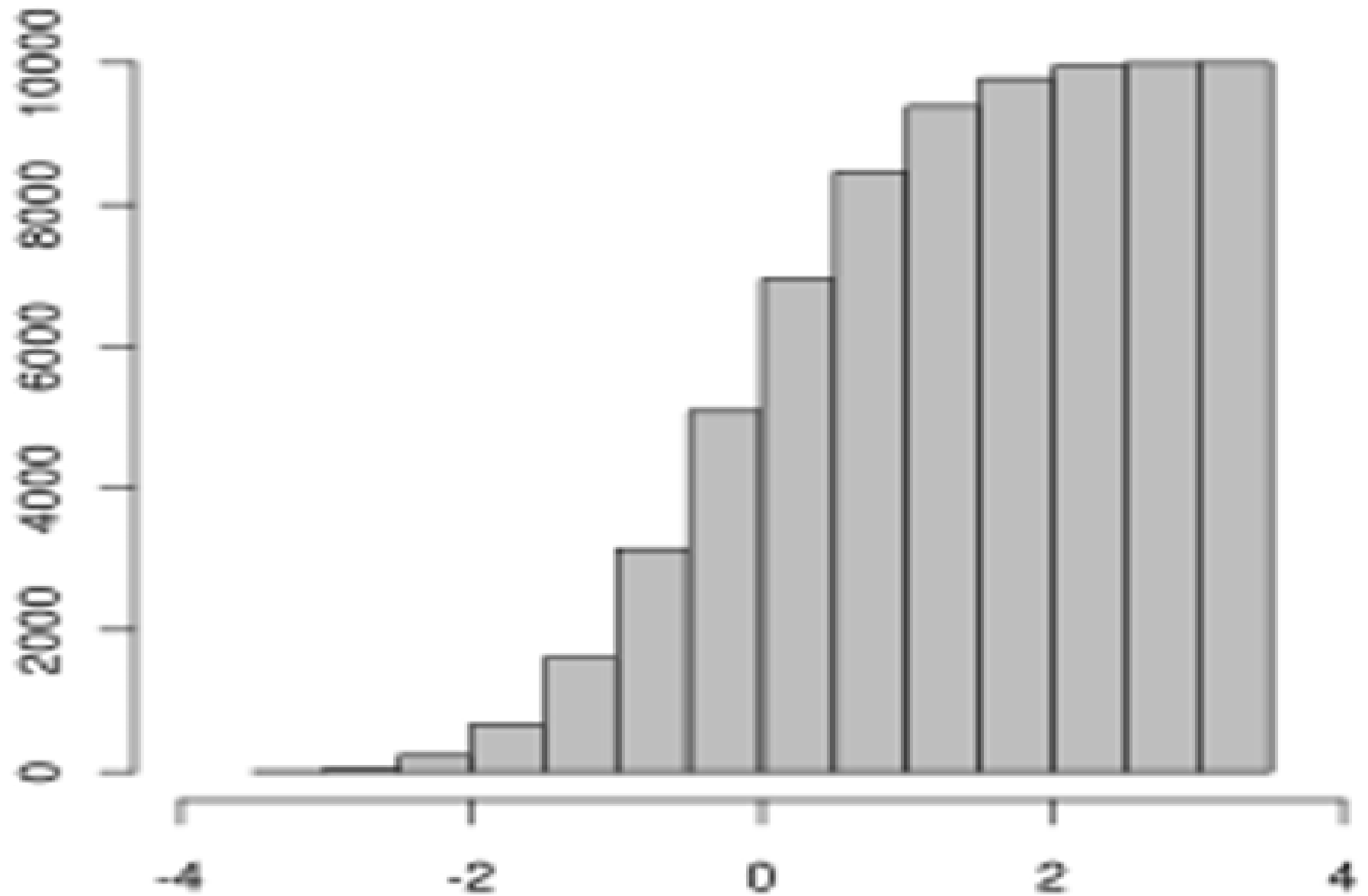
MCT (Symmetric and Skewed distributions)



MCT (Symmetric and Skewed distributions)

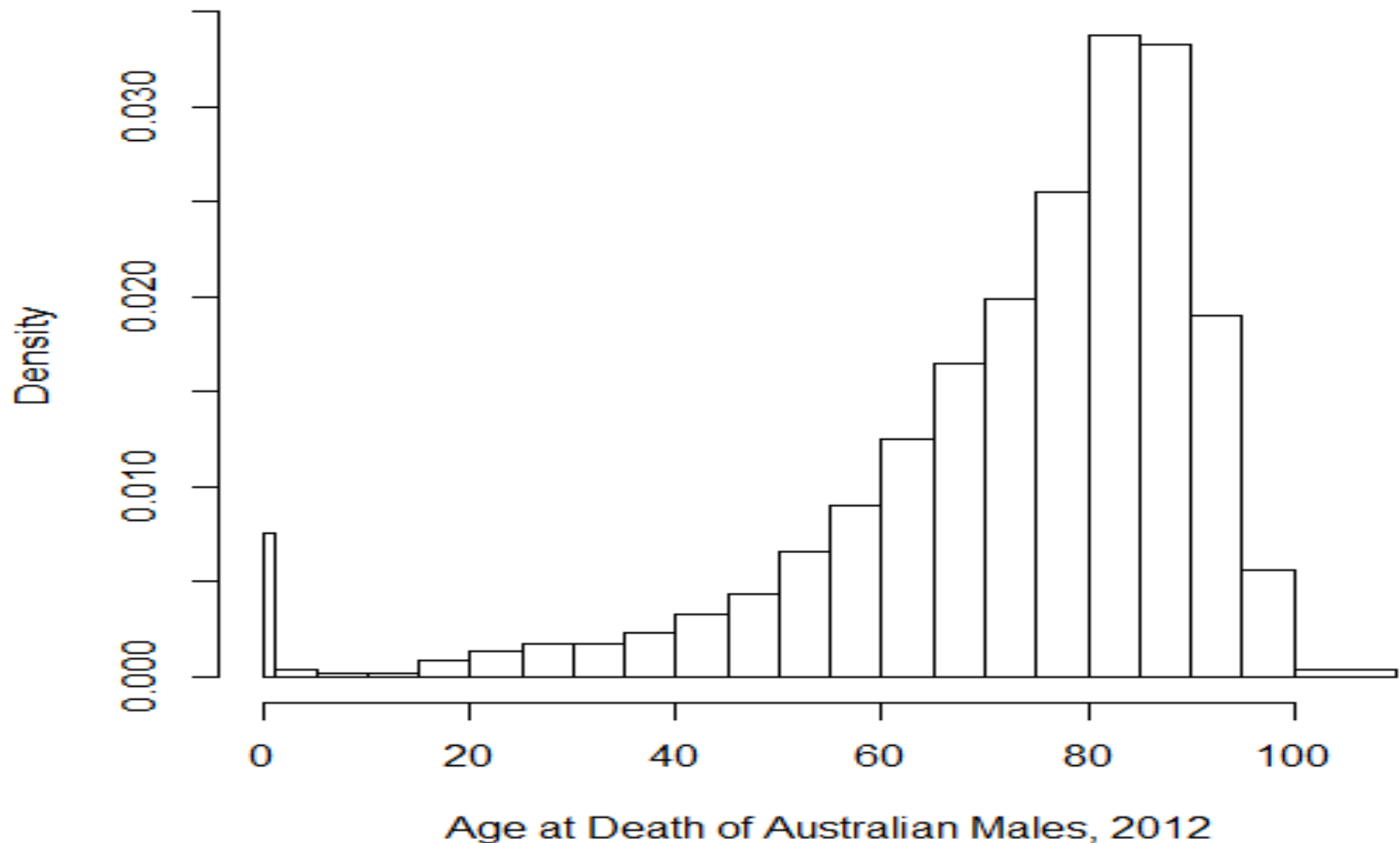


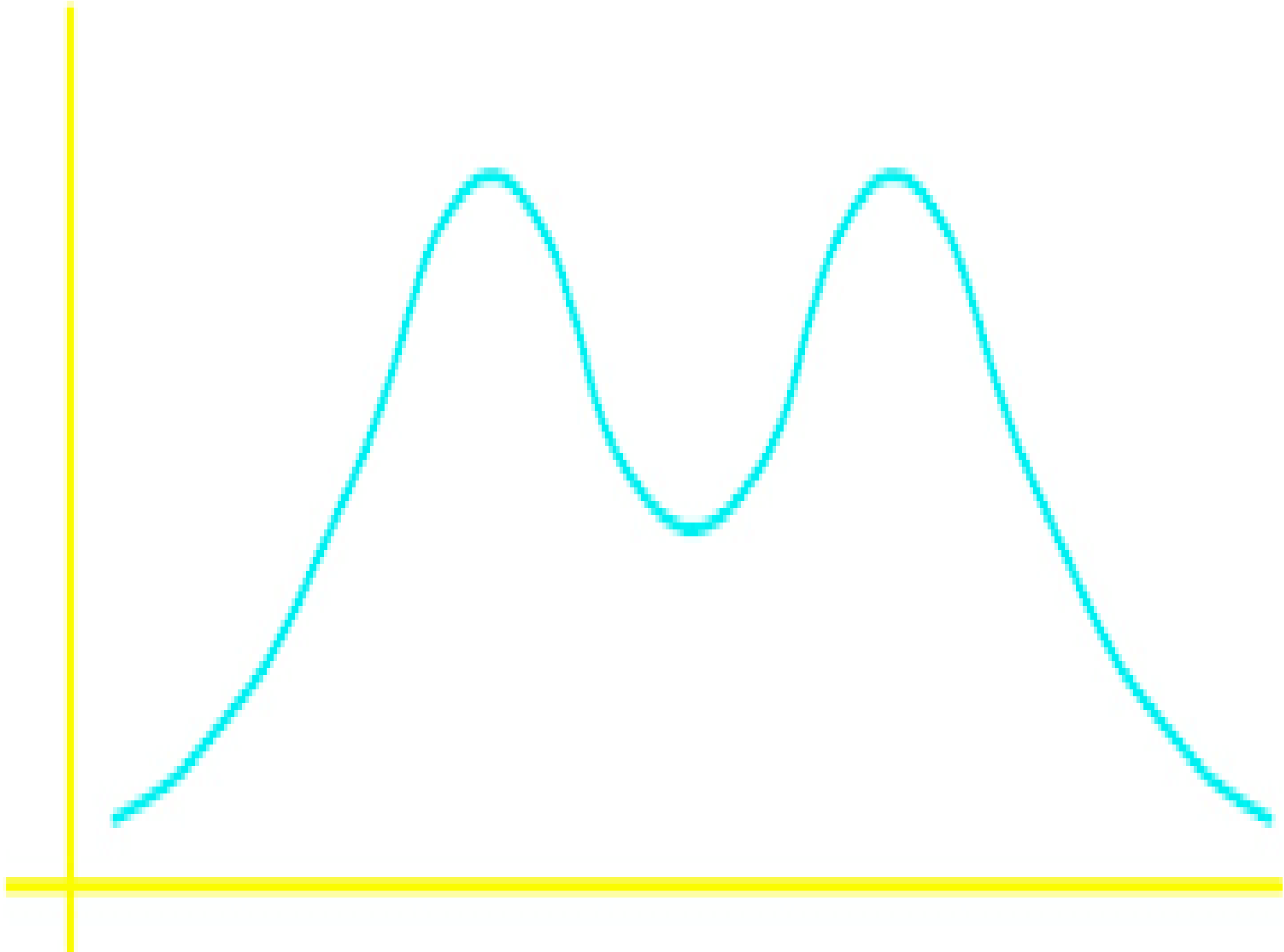
MCT (Symmetric and Skewed distributions)



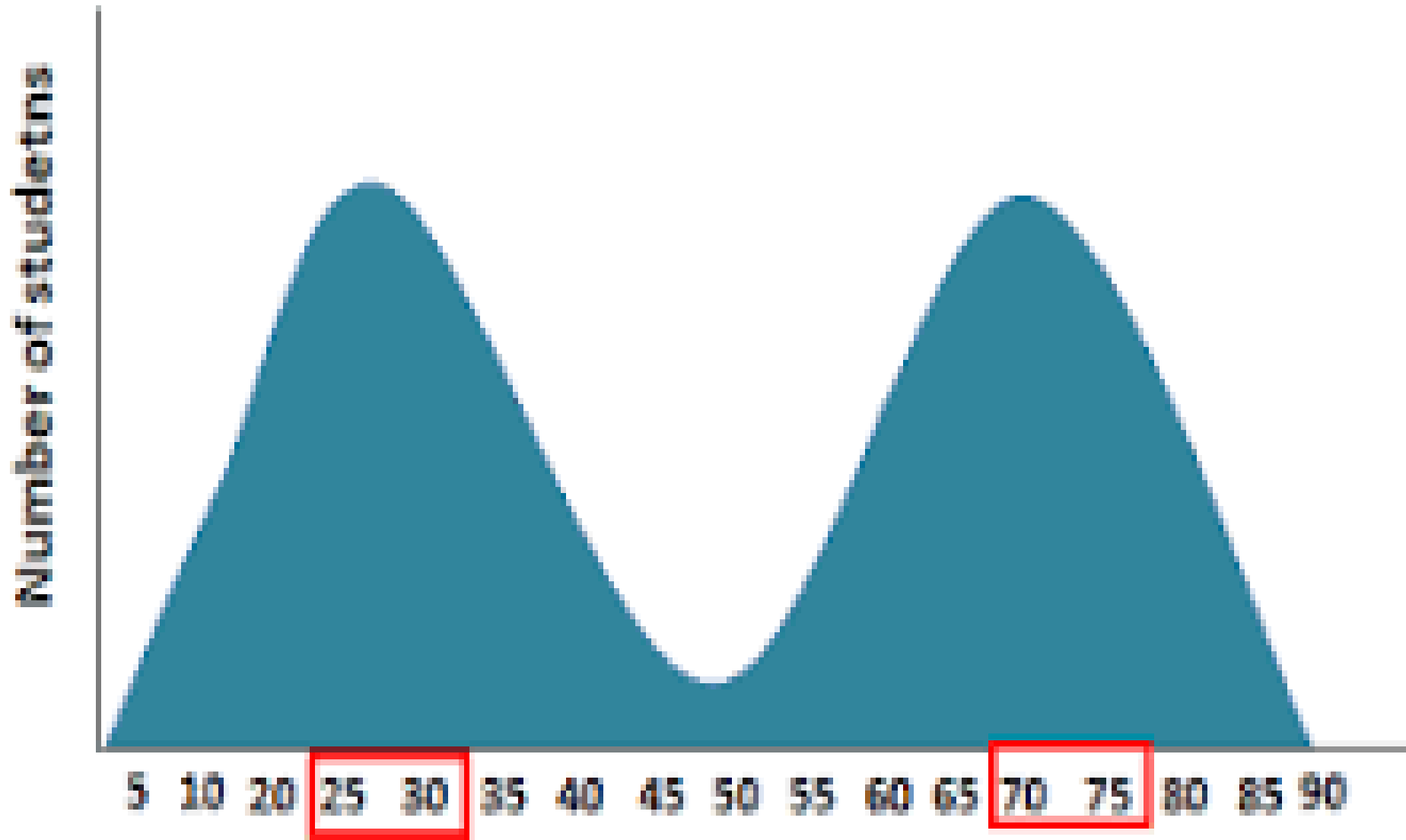
MCT (Symmetric and Skewed distributions)

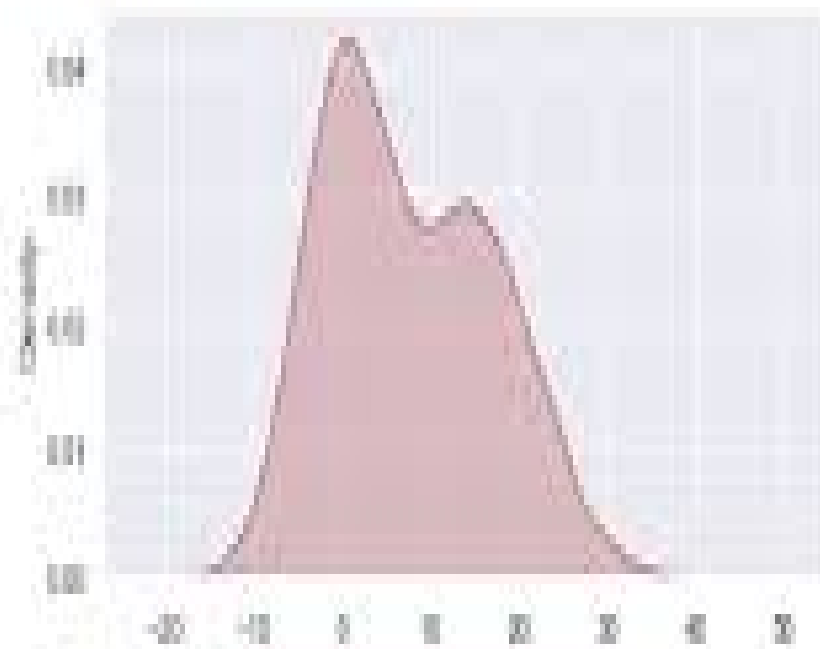
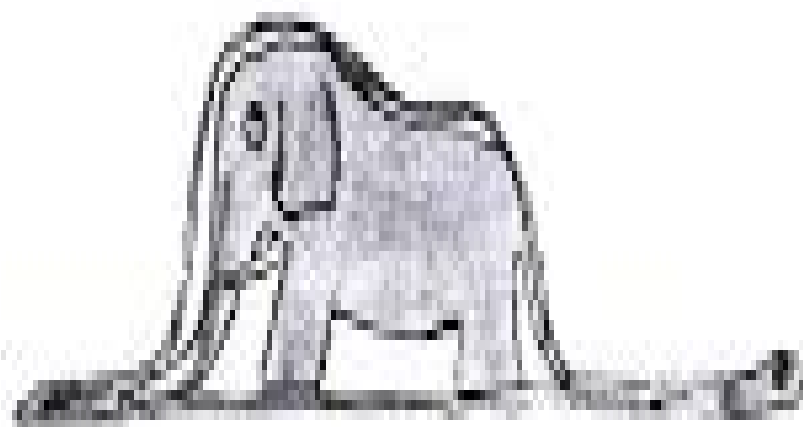
Histogram of Age at Death of Australian Males, 2012





MCT (Symmetric and Skewed distributions)





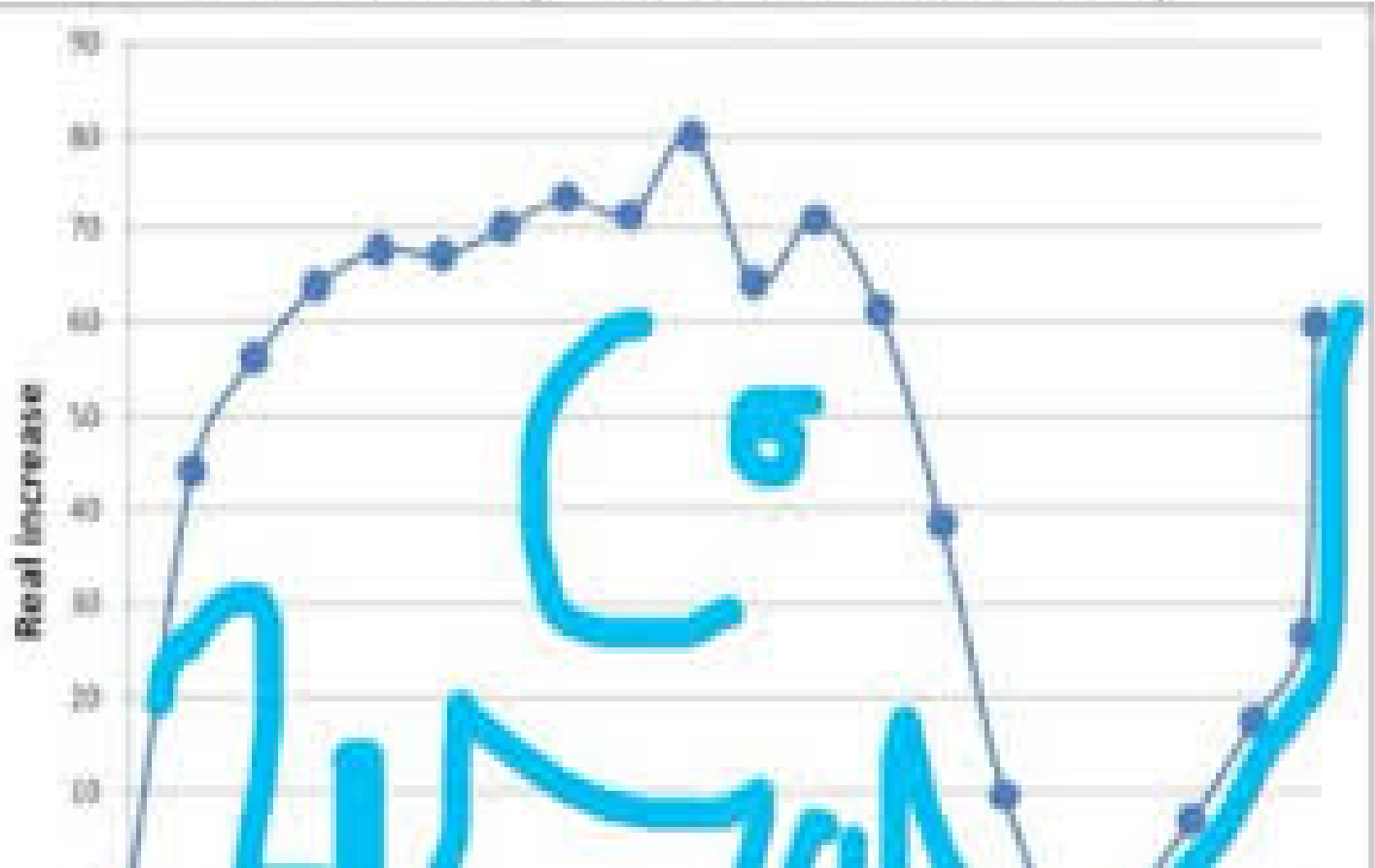
MCT (Symmetric and Skewed distributions)



Source: Author's adaptation of Milanovic, B. *Global Inequality*. Cambridge, MA: Harvard University Press, 2005; Figure 4.

MCT (Symmetric and Skewed distributions)

Figure 4. Change in real income between 1988 and 2008 at various percentiles of global income distribution (calculated in 2005 international dollars)

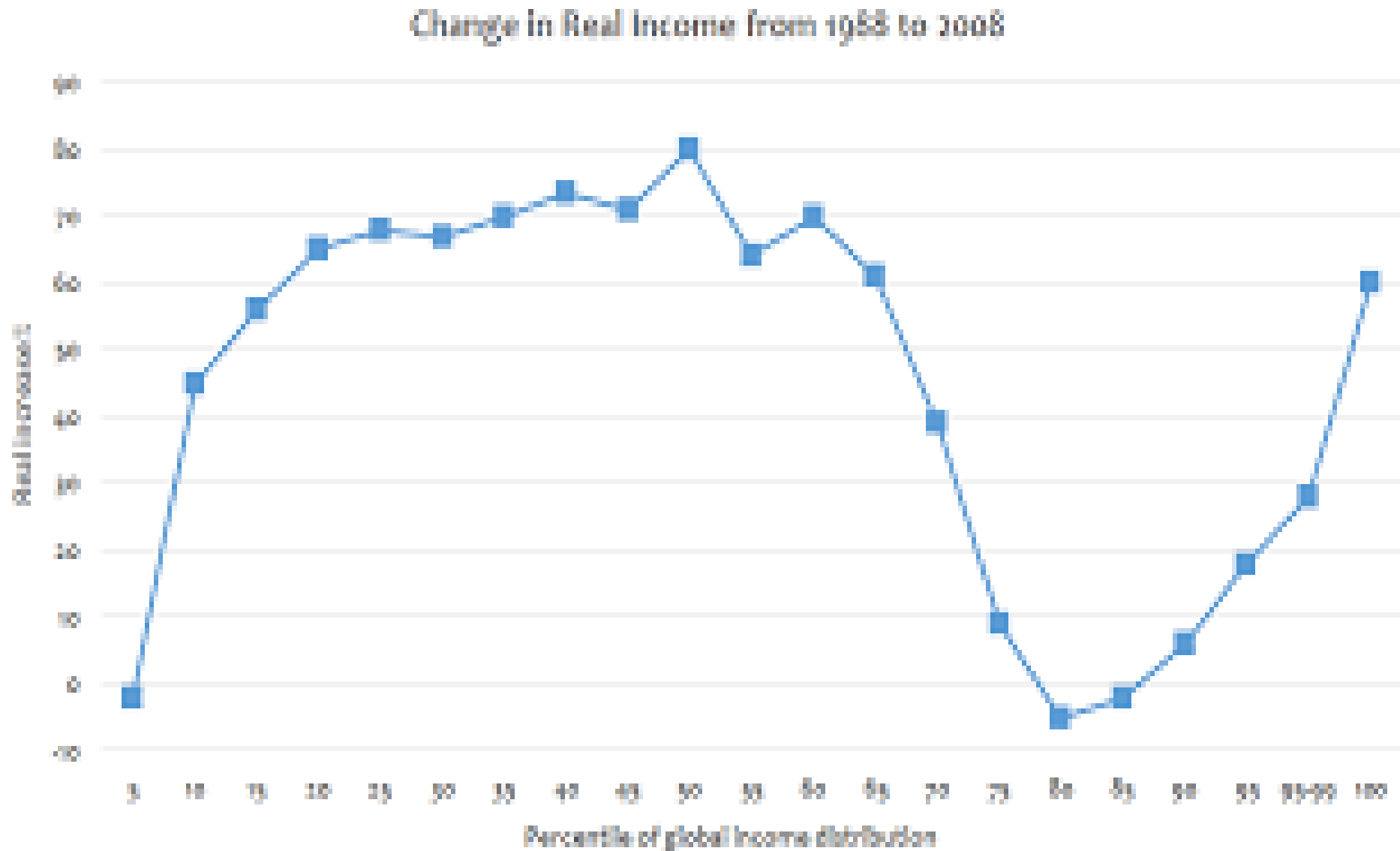


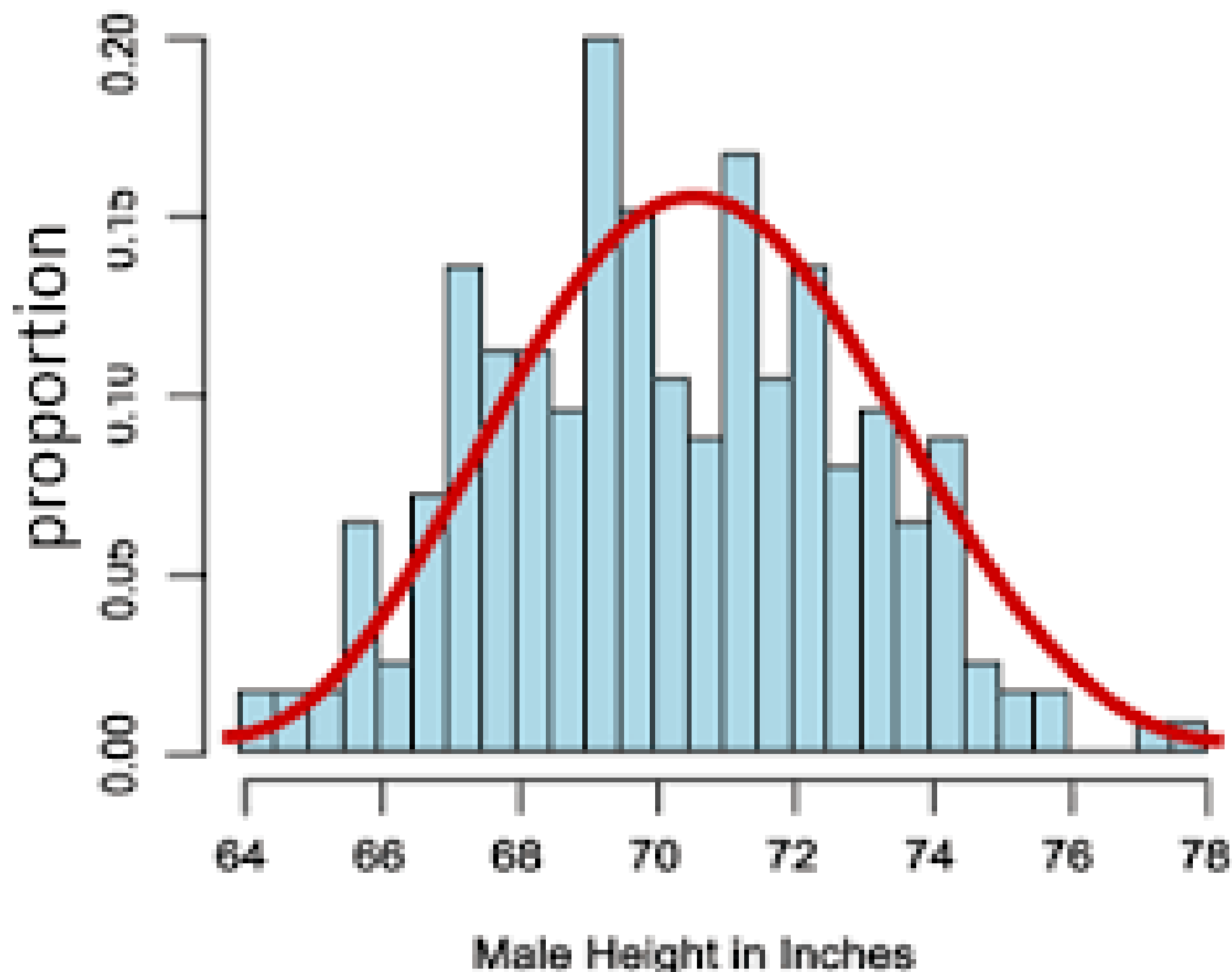
MCT (Symmetric and Skewed distributions)



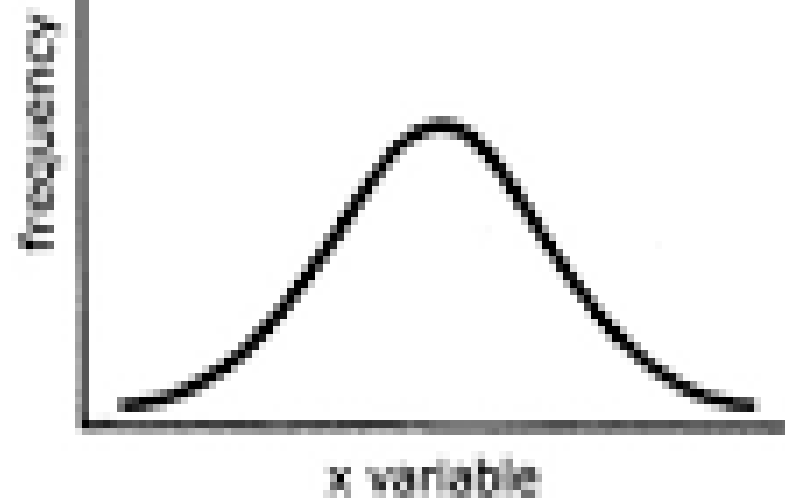
Elephant Chart

MCT (Symmetric and Skewed distributions)

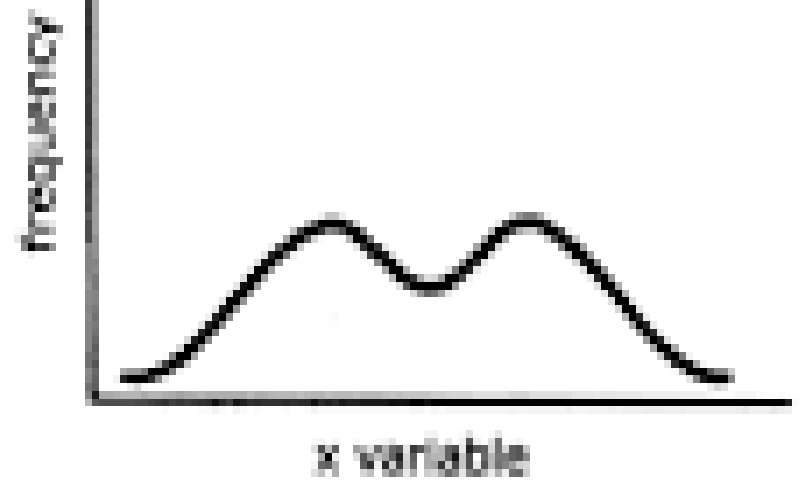




(1) Symmetric normal distribution



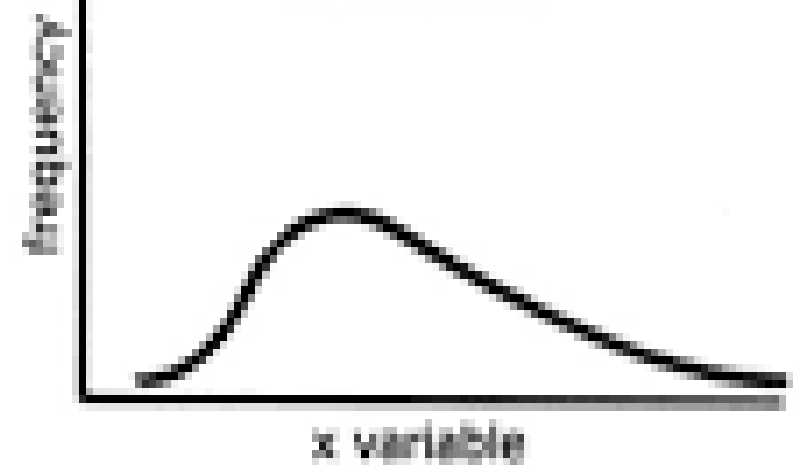
(2) Symmetric Bimodal distribution



(3) Negatively skewed distribution



(4) Positively skewed distribution

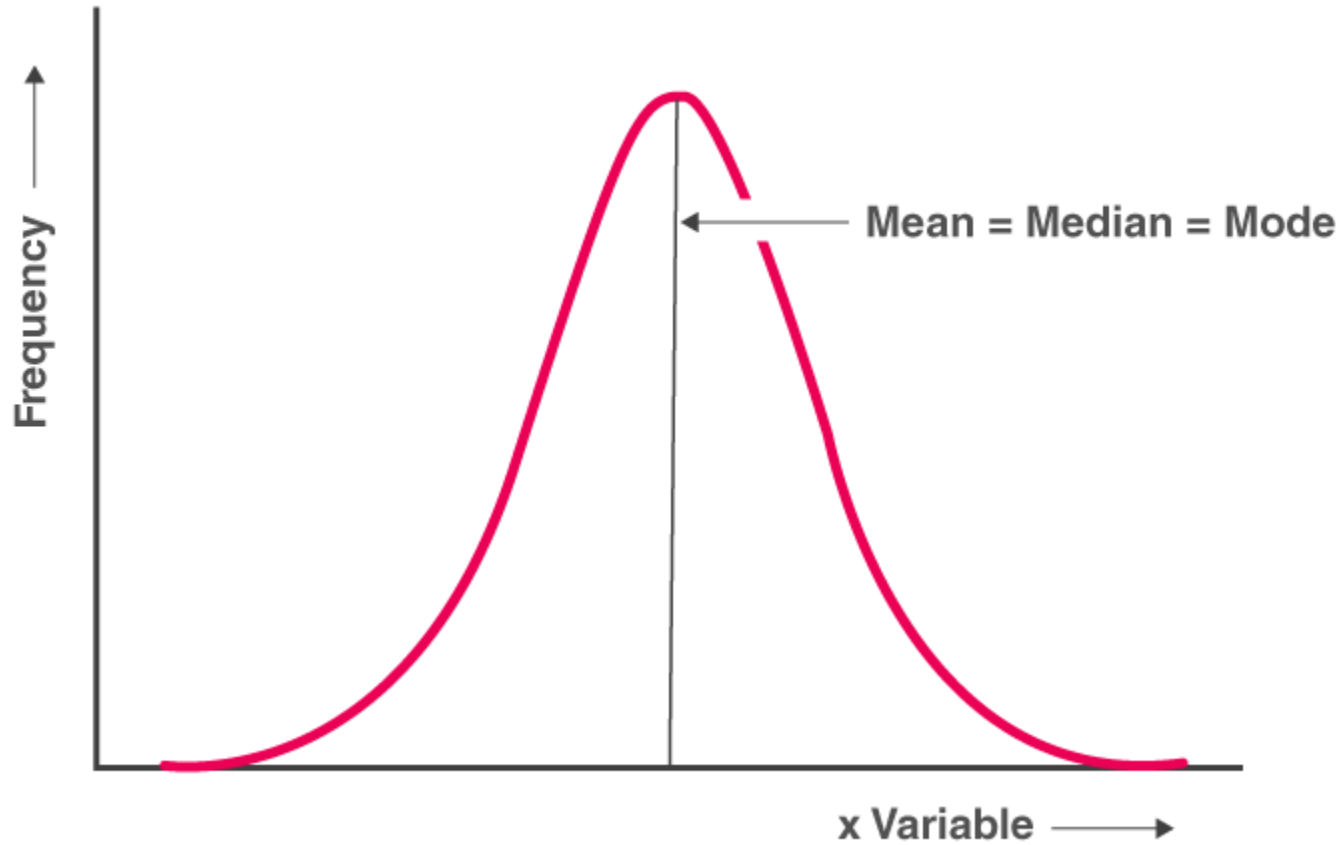


Relationships between Mean, Median and Mode

In statistics, for a moderately skewed distribution, there exists a relation between mean, median, and mode. This *mean median* and *mode* relationship is known as the “*empirical relationship*” which have been discussed in detail below.

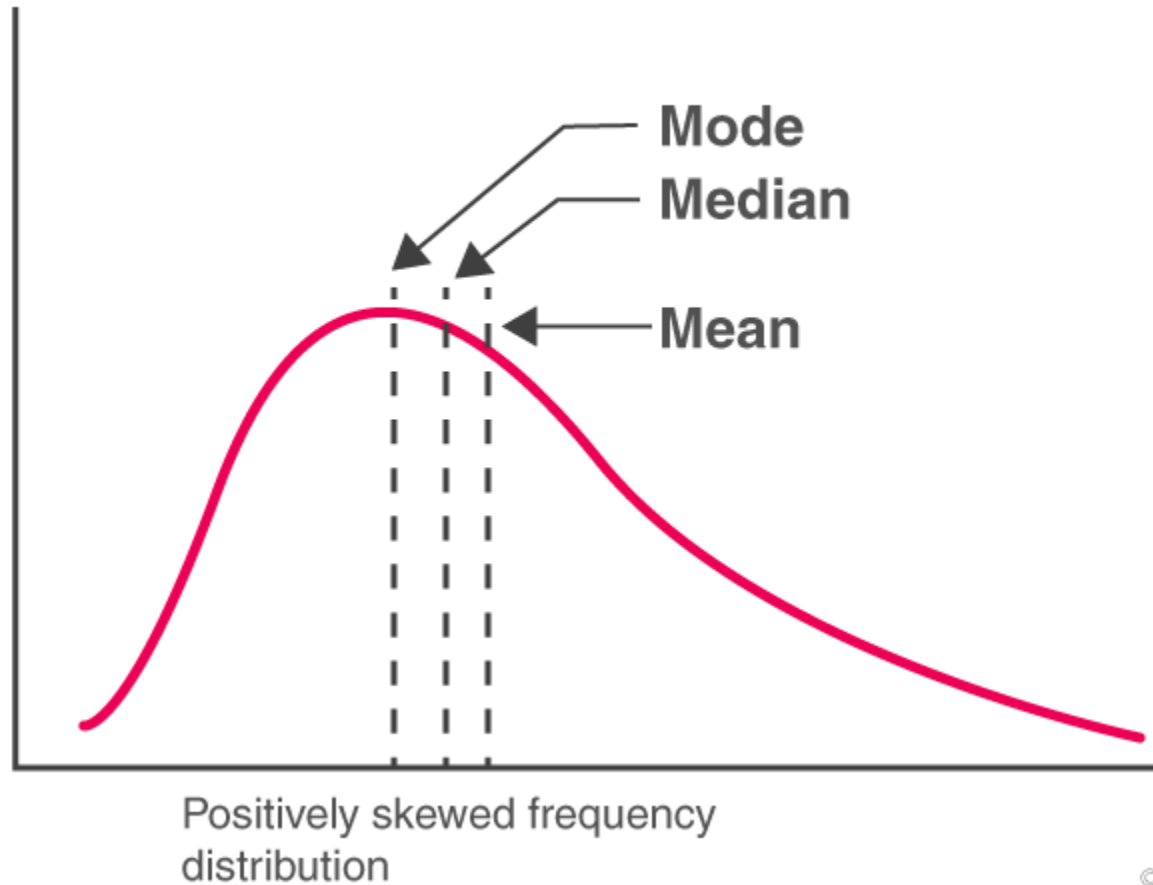
Empirical Relationship between Mean, Median and Mode

Mean = Median = Mode



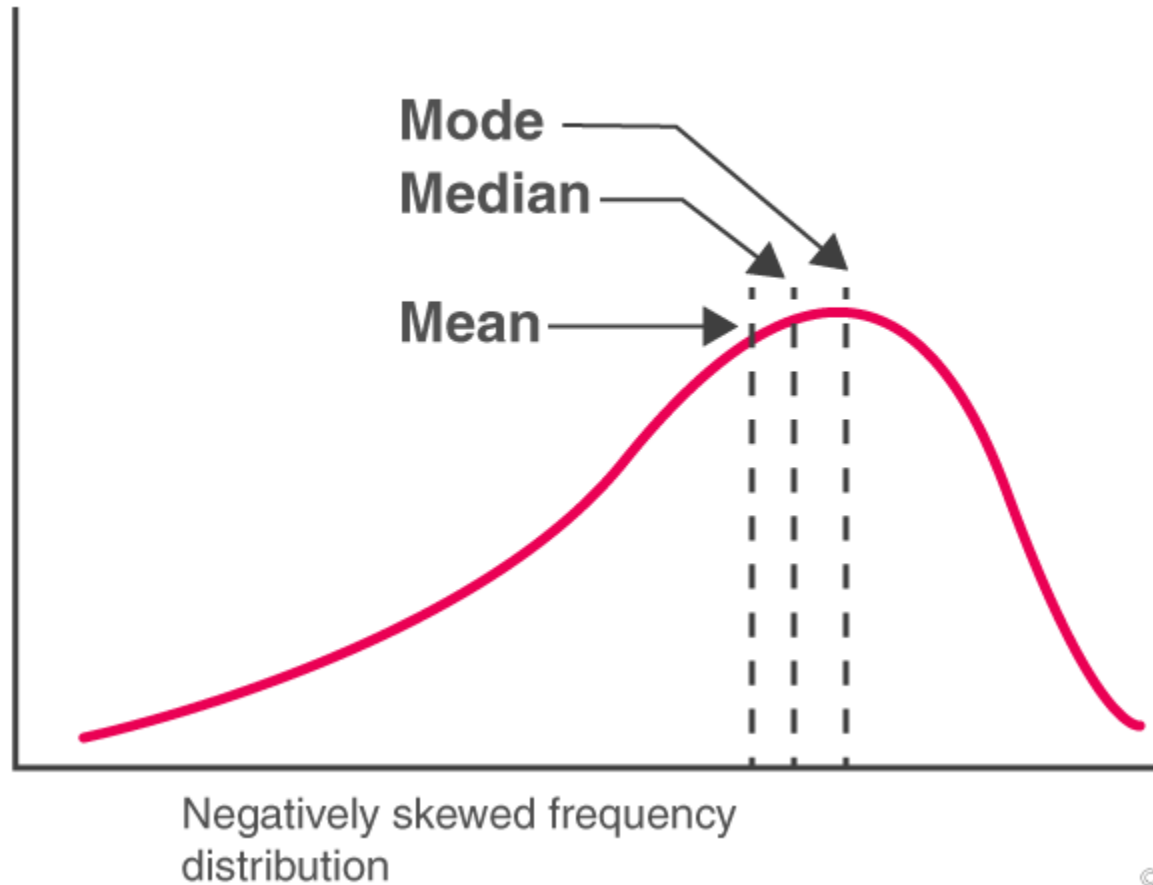
Empirical Relationship between Mean, Median and Mode

Mean > Median > Mode

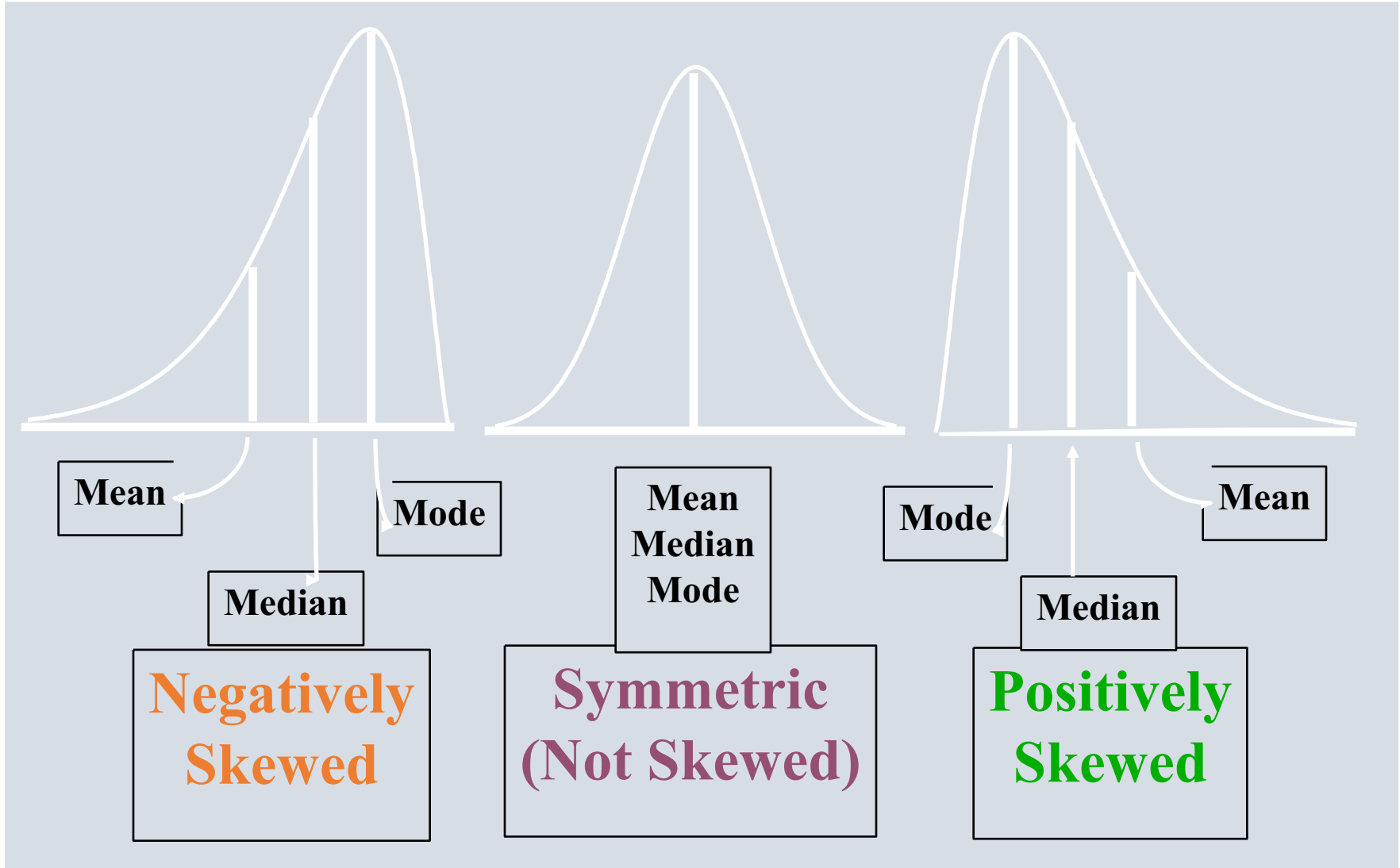


Empirical Relationship between Mean, Median and Mode

Mean < Median < Mode



Empirical Relationship between Mean, Median and Mode



Empirical Relationship between Mean, Median and Mode

In case of a moderately skewed distribution, the difference between mean and mode is almost equal to three times the difference between the mean and median. Thus, the *empirical mean median mode* relation is given as:

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

Empirical Relationship between Mean, Median and Mode

Example 1:- For a moderately skewed distribution,

the median is 55 and the mode is 51. Then

(i) the mean is _____

(ii) the type of skewness is _____.

Empirical Relationship between Mean, Median and Mode

Solution 1:- (i) $\tilde{x} = 55$ and $\hat{x} = 51$. Then

since $\bar{x} - \hat{x} = 3(\bar{x} - \tilde{x})$, it follows that

$$\bar{x} - 51 = 3(\bar{x} - 55)$$

$$\boxed{?} \bar{x} - 51 = 3\bar{x} - 165$$

$$\boxed{?} 165 - 51 = 3\bar{x} - \bar{x}$$

Empirical Relationship between Mean, Median and Mode

$$\Sigma x = 114 \quad \bar{x} = 57$$

(ii) Since $\bar{x} > \tilde{x} > \hat{x}$, the type of skewness is right or positive.

Empirical Relationship between Mean, Median and Mode

Example2:- In a moderately skewed distribution, the median is 20 and the mean is 22.5. Using these values:

- i) find the approximate value of the mode,
- ii) determine the type of skewness.

Empirical Relationship between Mean, Median and Mode

Example 3:- In a moderately asymmetrical distribution, the mode is twice the median and the mean is 37. Then the approximate value of the median is _____ the mode is _____ and the type of skewness is _____.

End of Chapter
4

Any Questions



Any Questions



Thank
you